

SANATH KAVATOORU

+1 (303) 414-0846 | sanathkavatooru@gmail.com | <https://linkedin.com/in/sanathkavatooru> | <https://github.com/sanath265>

Experience

Rule4

Boulder, CO

Software Engineer Intern – Capstone | NextJS, FastAPI, PostgreSQL, AWS, GCP

Oct 2025 – Jan 2026

- Spearheaded development of a full-stack resource scheduling platform from scratch, enabling preference-based allocation with points system, waitlisting, and auto-reallocation for **50+ premium resources**.
- Integrated **Google Calendar API** for two-way sync, enabling real-time availability updates and reducing scheduling conflicts by **40%**.
- Architected **RESTful APIs** with **FastAPI** and **PostgreSQL**, implementing role-based access control (RBAC), audit logging, and OWASP security best practices.
- Deployed production-ready application on **AWS RDS** and **S3** with **GCP** infrastructure, delivering a scalable enterprise solution adopted by the company.

AirAsia MOVE

Bangalore, Karnataka

Software Engineer | Swift, UIKit, MVVM, Combine

Feb 2022 – Jul 2024

- Developed a social profile screen in **UIKit** and refactored chat theming (fonts, wallpapers, colors), adopted by **60%** of monthly active users.
- Migrated chat UI to **MVVM** and consolidated **15** legacy screens into reusable components, reducing new UI development time by **35%**.
- Partnered with backend to migrate chat from a third-party provider to an in-house stack, saving **\$12K/year** and improving product control.
- Replaced Promise-based async flows with **Combine** pipelines to improve cancellation/retry handling and make networking/UI state updates more resilient.
- Added **50+ XCUITest** UI tests running on every build and nightly with CI reports, catching regressions earlier and speeding **TestFlight** releases.
- Extended a Ruby CI pipeline integrated with **Git** to publish real-time pass/fail dashboards and artifacts, cutting regression triage time by **50%**.
- Engineered a streaming GIF renderer (replacing SDWebImage GIF handling), reducing peak memory per chat thread by **65%** via Xcode Instruments.

Projects

URL Shortener | ReactJS, Django, PostgreSQL, Docker, Kubernetes, GKE

- Launched a full-stack URL shortener using **ReactJS** and **Django** on **GKE**, supporting branded links, custom aliases, expiry, password protection, and geo-based rules.
- Deployed containerized backend on **Google Cloud** with **Docker** and **Kubernetes**, achieving **99.9%** uptime and handling **10K+ redirects/day**.

AI Learning Assistant | Python, Streamlit, LangChain, ChromaDB, RAG, STT/TTS

- Created an AI learning assistant using **Python** and **Streamlit** for quizzes, document-based study, and mock interviews, serving **100+ study sessions**.
- Implemented a multimodal **RAG pipeline** ingesting **50+ PDFs**, generating embeddings in **ChromaDB**, achieving **85%** answer accuracy on grounded responses.
- Integrated **STT/TTS** for voice-first interaction with **<500ms** latency, and orchestrated **LangChain** agents for adaptive coaching across difficulty levels.

AudioTranscriptor | Swift, AVAudioEngine, Google Cloud, OpenAPI

- Crafted an iOS app in **Swift** using **AVAudioEngine** to record sessions up to **2+ hours**, chunking audio into **30s clips** with **99%** upload success rate.
- Designed a cloud upload pipeline to **Google Cloud**, processing **500+ audio files** with resumable uploads, retries, and backpressure handling.
- Shipped a transcript playback UX with **<100ms** text-audio sync latency, tap-to-seek, and **60%** faster navigation for long recordings.

Skills

Frontend: React, NextJS, TypeScript, JavaScript, Tailwind CSS, HTML/CSS, Streamlit

Backend & APIs: Python, FastAPI, Django, Node.js, NestJS, REST, JWT, WebSockets

Cloud & DevOps: AWS (EC2, Lambda, RDS, S3), Google Cloud (GCS, GKE), Docker, Kubernetes, Git, CI/CD

Data & Storage: PostgreSQL, MySQL, MongoDB, Redis, ChromaDB

AI & ML: OpenAI API, RAG, LangChain, Embeddings, Prompt Engineering, STT/TTS

Languages: Python, JavaScript, TypeScript, SQL, C++, Bash

Education

University of Colorado Boulder

May 2026

Master of Science, Computer Science

CGPA: 3.84/4